

Richard Suhendra

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EDUCATION

- **Northwestern University** Evanston, IL
PhD in Applied Mathematics | GPA 3.89/4.0 2020 – 2026
- **The Pennsylvania State University** State College, PA
BS in Mathematics (Honors), Minor in Statistics | GPA 3.88/4.0 2016 – 2020

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, Bash, SQL, R, MATLAB, Java
- **Machine Learning & AI:** scikit-learn, PyTorch, XGBoost, HuggingFace Transformers, LangChain, spaCy, statsmodels, OpenCV
- **Data & Workflow:** pandas, SQL, PySpark, Snowflake, PostgreSQL, AWS, Docker, Git
- **Selected Skills:** Machine Learning, Statistical Modeling, Credit Risk & Financial Modeling, Stochastic Modeling & Simulation, Time-Series Analysis, Optimization, Causal Inference, High-Performance Computing, Natural Language Processing

EXPERIENCE

- **Research Scientist Intern** 2025
Upstart (Machine Learning Department)
 - Built production-ready loan delinquency models on 500M+ records using neural networks and gradient-boosted trees for large-scale default risk prediction.
 - Developed a Markov chain-based forecasting model that chained XGBoost-estimated next-day delinquency probabilities to predict long-term default risk.
 - Improved default prediction over baseline models, increasing AUC by 4% and reducing log loss by 5%; validated results through calibration metrics and leakage-aware backtesting.
- **Graduate Teaching Assistant** 2024 – 2026
Northwestern University
 - Designed and led hands-on NLP and machine learning labs, guiding implementations of transformers, RAG systems, and PyTorch-based deep learning workflows.
 - Revamped course infrastructure by migrating MATLAB assignments to Python, adding unit tests, and building a GitHub Actions auto-grader that automatically evaluated submissions.

RESEARCH PROJECTS

- **A model for goal-directed navigation in a thermal gradient** 2023 – Present
Department of Engineering Sciences and Applied Mathematics, Northwestern University
 - Designed a mathematical model of temperature-guided navigation, linking behavioral data to candidate neural circuit mechanisms.
 - Adapted stochastic differential equation (SDE) models to fit behavioral time series, applying gradient-free optimization and custom metrics to select the best model.

- Engineered OpenCV-based computer vision pipelines to track fly movement trajectories and applied time-series analysis to uncover behavioral patterns.
- Validated results using generalized linear mixed models (GLMMs) and nonparametric hypothesis testing to demonstrate significant differences in treatment.
- **Mechanisms of adaptation to extreme temperatures in the snow fly** 2023 – 2025
Department of Neurobiology, Northwestern University
 - Built an end-to-end bioinformatics pipeline that transformed multimodal genomic data into protein sequences for downstream annotation and comparative analysis.
 - Fine-tuned Meta ESM2 protein LLM in PyTorch to identify antifreeze proteins ($F_1=0.80$), using model predictions to prioritize candidate proteins for biological interpretation.
 - Applied numerical PDEs (Finite Element Method) to model spatiotemporal temperature dynamics within experimental assays.
- **Rapid threat assessment in the *Drosophila* thermosensory system** 2022 – 2023
Department of Engineering Sciences and Applied Mathematics, Northwestern University
 - Utilized Monte Carlo simulation to match stochastic agent-based models to fly movement patterns, integrating a neuronal derivative model to achieve a 20% reduction in overall error.
 - Developed multi-objective genetic algorithms in DEAP to optimize model parameters, utilizing a high-performance computing (HPC) environment to accelerate computations by 40x.
- **Determining causal structure with Bayesian networks** 2022
Data-Driven Dynamical Systems Final Project, Northwestern University
 - Implemented predictive modeling of non-stationary biological time series data using time-varying dynamic Bayesian networks.
 - Demonstrated that method outperformed traditional auto-regressive (AR) models in capturing high-dimensional dependencies and temporal variability, increasing F_1 scores by up to 15%.
- **Speeds of biological invasion** 2018 – 2020
Department of Mathematics, The Pennsylvania State University
 - Analyzed ecological invasions by deriving bounds on invasion speeds in a stochastic model using mean-field reaction-diffusion theory, providing insights into population spreading.
 - Implemented a lattice-free 2D stochastic invasion model, and used computational simulations to validate its alignment with KPZ stochastic PDE theory.

PUBLICATIONS

1. Capek M, **Suhendra R**, ... , Kath WL, Stensmyr M, Gallio M (2026). *Coordinated molecular and physiological adaptations enable activity at sub-freezing temperatures in the snow fly *Chionea alexandriana**. Current Biology.
2. Jouandet GC, Alpert MH, Simões JM, **Suhendra R**, ... , Kath WL, Gallio M (2023). *Rapid threat assessment in the *Drosophila* thermosensory system*. Nature Communications.
3. **Suhendra R** (2020). *Speeds of Biological Invasion*. Honors Thesis.
4. **Suhendra R**, Simões JM, ... , Kath WL, Gallio M. *A model for goal-directed navigation in a thermal gradient based on *Drosophila* behavior*. Expected submission: 2026